TUTORIAL 5 INTRODUCTION TO MIPS ASSEMBLY

Overview

- **You will review/learn the following in this tutorial:**
  - Instruction Set Architecture (ISA)
  - MIPS data types
  - Pseudo instructions
- **You will practice to write short MIPS programs:**
  - Understand the flow of programs and describe their behaviour

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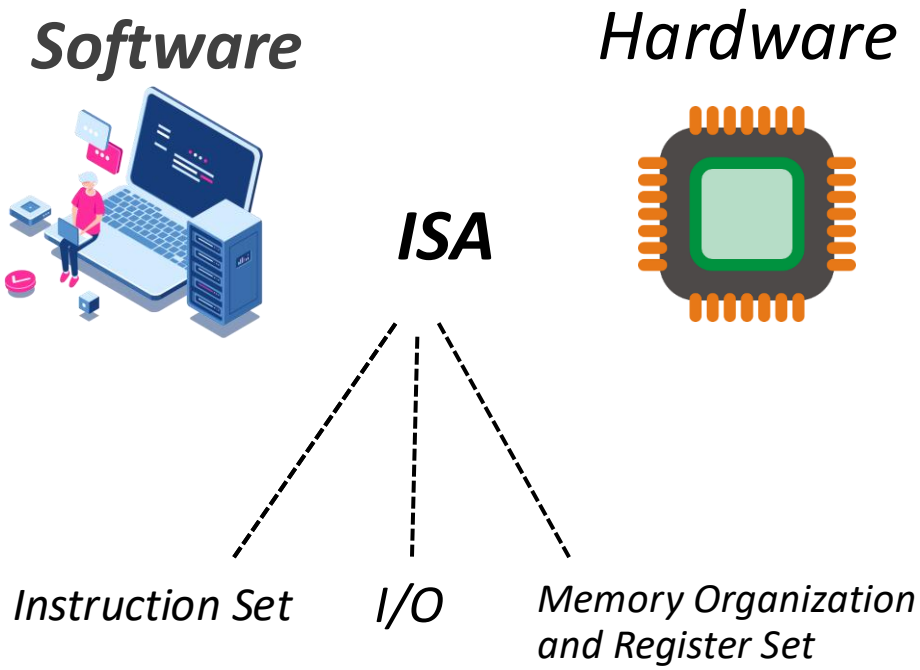
Instruction Set Architecture

- **ISA defines the hardware/software interface**



ISA Types:

- *CISC (Complex Instruction Set Computer)* *e.g., x86*
- *RISC (Reduced Instruction Set Computer)* *e.g., ARM, MIPS*

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Instruction Set Architecture (cont.)

- **instruction set architecture (ISA), is the part of the processor that is visible to the programmer or compiler writer**
  - the native data types,
  - instructions,
  - registers,
  - addressing modes,
  - memory architecture,
  - interrupt and exception handling,
  - and external I/O.

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MIPS Registers

- **The MIPS central processing unit contains**
  - 32 general purpose 32-bit registers
  - numbered 0-31
- **MIPS has established a set of conventions as to how registers should be used**
  - not enforced by the hardware

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MIPS General Purpose Registers

Register number	*Name*	*Used for*
0	`zero`	Always returns 0
1	`at`	(assembly temporary) Reserved for use by assembly
2–3	`v0`, `v1`	Value returned by subroutine
4–7	`a0–a3`	(arguments) First few parameters for a subroutine
8–15	`t0–t7`	(temporaries) Subroutines can use without saving
24, 25	`t8`, `t9`	
16–23	`s0–s7`	Subroutine register variables; a subroutine that writes one of these must save the old value and restore it before it exits, so the *calling* routine sees the values preserved
26, 27	`k0`, `k1`	Reserved for use by interrupt/trap handler; may change under your feet
28	`gp`	Global pointer; some runtime systems maintain this to give easy access to (some) `static` or `extern` variables
29	`sp`	Stack pointer
30	`s8/fp`	Ninth register variable; subroutines that need one can use this as a frame pointer
31	`ra`	Return address for subroutine

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MIPS Data Types

.ascii	String (without null terminator)
.asciiz	String (with null terminator)
.byte	Byte (We can write the values either in base 10 or hex)
.half	2 Bytes (We can write the values either in base 10 or hex)
.word	4 Bytes (We can write the values either in base 10 or hex)
.space num	Reserves num bytes of space in memory.

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Examples

Variable name	Data type	Initialized value	Remarks
var1:	.half	14	# A half-word storing the integer 14
array1:	.word	5 6 7 8	# same as int array1[4] = {5,6,7,8} in C++
array2:	.word	3:5	# the part before ":" in the initialized value is # the initial value of each element in the # array, and the part after ":" is the array size. # same as int array2[5] = {3,3,3,3,3} in C++
string1:	.byte	0x32 # '2' in ASCII code 0x4a # 'J' in ASCII code 0 # '\0' in ASCII code	# string type is actually an array of char (a byte) # same as char string1[3] = {'2','J','\0'} in C++
string2:	.asciiz	"2J"	# equivalent to string1
array3:	.space	10	An array of 10 bytes is allocated for array3 in memory.

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Warm up exercise 1

- **Write down the shortest sequence (any one) of MIPS instructions for the following C++ code, assuming variable a is stored in $s0**
  - a = a - 1;

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Warm up exercise 1

- **Write down the shortest sequence (any one) of MIPS instructions for the following C++ code, assuming variable a is stored in $s0**
  - a = a - 1;

Solution:

- **addi $s0, $s0, -1**
- **# note: no subi instruction in MIPS (it is pseudo-instruction)**

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Warm up exercise 2

- **Write down the shortest sequence (any one) of MIPS instructions for the following C++ code, assuming variable a,b are stored in $s0 and $s1 respectively**
  - b = a * 5;

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Warm up exercise 2

- **Write down the shortest sequence (any one) of MIPS instructions for the following C++ code, assuming variable a,b are stored in $s0 and $s1 respectively**
  - b = a * 5;

Solution:

- **sll $s1, $s0, 2**
- **add $s1, $s1, $s0**

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MIPS Hello World

HelloWorld.asm | AddSub.asm | Logical.asm | Shift.asm | Pseudo.asm | Memory.asm

```
#helloworld
.data #data segment
message: .asciiz "\nHello World!\n"

.text #text segment
.globl _main
_main: #your code starts here
li $v0, 4 #load syscall code 4 to $v0 for print string
la $a0, message #load address of string to $a0
syscall #invoke syscall
li $v0, 10 #load syscall code 10 to $v0 for exit
syscall #invoke syscall to terminate the program
```

Mars Messages | Run I/O

Clear

```
Hello World!

-- program is finished running --
```

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MIPS Arithmetic Operation

HelloWorld.asm | AddSub.asm | Logical.asm | Shift.asm | Pseudo.asm | Memory.asm

```
#simple arithmetic
.data # data segment
.text # text segment
.globl _main
_main: #your code starts here
sub $t0, $t0, $t0 #init t0=0
addi $t1, $zero, 1 #init $t1=1
addi $t2, $0, -2 #init $t2=-2
add $t3, $t0, $t1
add $t3, $t3, $t2 #t3=t0+t1+t2
```

$t0	8	0x00000000
$t1	9	0x00000001
$t2	10	0xfffffffe
$t3	11	0xffffffff
$t4	12	0x00000000

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MIPS Logical Operation

HelloWorld.asm | AddSub.asm | Logical.asm | Shift.asm | Pseudo.asm | Memory.asm

```
#logical operations
.data # data segment
.text # text segment
.globl _main
_main: #your code starts here
addi $s0, $0, 0xffff0000
addi $s1, $0, 0xaaaa1111
not $t0, $s0
and $t1, $s0, $s1
or $t2, $s0, $s1
nor $t3, $s0, $s1
xor $t4, $s0, $s1
```

$t0	8	0x0000ffff
$t1	9	0xaaaa0000
$t2	10	0xffff1111
$t3	11	0x0000eeee
$t4	12	0x55551111

MIPS Logical Shift

HelloWorld.asm | AddSub.asm | Logical.asm | Shift.asm* | Pseudo.asm | Memory.asm

```
_main: #your code starts here
addi $t0, $zero, 0xf0
sll $t1, $t0, 4 #logical shift
srl $t2, $t0, 4

addi $t0, $zero, 0xffff0000
srl $t1, $t0, 4
sra $t2, $t0, 4 #arithmetic shift

addi $t0, $0, 1 #t0=1
sll $t0, $t0, 3 #t0=8*t0, a trick for fast multiplication

addi $t0, $0, 0x12345678
addi $t1, $0, 0xf
sll $t1, $t1, 4
and $t1, $t0, $t1 #pick bit 4-7 of $t0
```

Pseudo-instruction

- The MIPS (real) instruction set is very small
- **Pseudo-instructions** are assembly language instructions that do not have a direct hardware implementation
  - Defined by assembler
  - When using pseudo-instructions in an assembly language program, the assembler translates them into equivalent real MIPS instructions
  - Provided as a convenience for the programmer

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MIPS Pseudo-instructions

- abs (absolute value)
- blt (branch if less than)
- bgt (branch if greater than)
- ble (branch if less than or equal)
- bge (branch if greater than or equal)
- neg (computes the two's complement negative value)
- not (bitwise ’flip’, 1 is changed to 0 and 0 is changed to 1)
- li (load immediate)
- la (load address)
- move (move the contents of one register to another)
- sge (set greater than equal)
- sgt (set greater than)

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Example

HelloWorld.asm | AddSub.asm | Logical.asm | Shift.asm | Pseudo.asm* | Memory.asm

```
#Pseudo instructions
.data # data segment
message: .asciiz "\nPseudo instructions!\n"
.text # text segment
.globl _main
_main: #your code starts here
subi $t0, $zero, 1 #t0=-1
move $t1, $t0 #t1=t0
li $v0, 4 #load syscall code 4 to $v0 for print strin
la $a0, message #load address of string to $a0
syscall #invoke syscall
li $v0, 1 #load syscall code 4 to $v0 for print
move $a0, $t1 #load address of string to $a0
syscall #invoke syscall
```

Data Transfer

Memory.asm

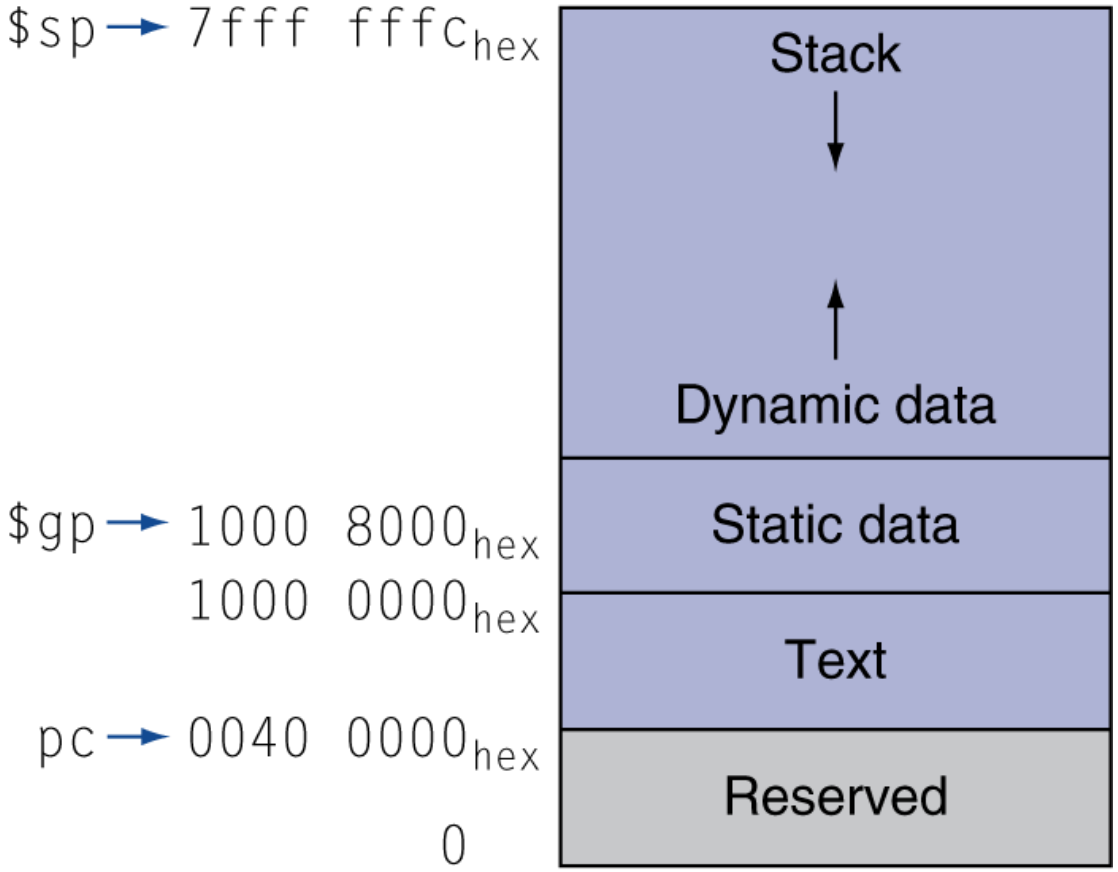
```
# data transfer
.data # data segment
fourbytes: .ascii "12AB"
fourwords: .word 1 -1 1024 -65536
.text # text segment
.globl _main
_main: #your code starts here
# let's play with byte array first
la $s0, fourbytes #base address of byte array
lb $t0, 0($s0)
lb $t1, 2($s0)
```

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)
0x10010000	0x42413231	0x00000001	0xffffffff	0x00000400	0xffff0000

$s0	16	0x10010000
$s1	17	0x10010004

MIPS Memory Layout



Load Byte

```
# let's play with byte array first
la $s0, fourbytes #base address of byte array
lb $t0, 0($s0)
lb $t1, 2($s0)
lb $t2, 1($s0)
lb $t3, 3($s0)
```

$t0	8	0x00000031
$t1	9	0x00000041
$t2	10	0x00000032
$t3	11	0x00000042

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Load Word

```
# check the word array
la $s1, fourwords
lw $t0, 0($s1)
lw $t1, 4($s1)
lw $t2, 8($s1)
lw $t3, 12($s1)
```

$t0	8	0x00000001
$t1	9	0xffffffff
$t2	10	0x00000400
$t3	11	0xffff0000

Exercise

- **Write down the shortest sequence (any one) of MIPS instructions for the following C++ code, assuming the base address of the int array A is stored in the register s0. You can use $t0 for storing temporary values.**
  - A[2] = A[7] + 11;

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Exercise

- **Write down the shortest sequence (any one) of MIPS instructions for the following C++ code, assuming the base address of the int array A is stored in the register s0. You can use $t0 for storing temporary values.**
  - A[2] = A[7] + 11;

Solution:

- **lw $t0, 28($s0)**
- **addi $t0, $t0, 11**
- **sw $t0, 8($s0)**

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Extra exercise

- **Write down the shortest sequence (any one) of MIPS instructions for the following C++ code, assuming the base address of the integer array A, variable b, variable c are stored in the register $s0, $s1 and $s2 respectively. You can use some registers for storing temporary values.**
  - A[c++] = A[b] + 17;

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Extra exercise

Solution:

- sll $t0, $s1, 2 # $s1 stores b
- add $t0, $s0, $t0
- lw $t1, 0($t0)
- addi $t1, $t1, 17
- sll $t0, $s2, 2 # $s2 stores c
- add $t0, $s0, $t0
- sw $t1, 0($t0)
- addi $s2, $s2, 1

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